

Glycosylated IgG antibodies accelerated the recovery of haemorrhagic fever with renal syndrome patients

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eLife Assessment

The authors investigated the potential role of IgG N-glycosylation in Haemorrhagic Fever with Renal Syndrome (HFRS), which may offer significant insights for understanding molecular mechanisms and for the development of therapeutic strategies for this infectious disease. The findings are thought to be **valuable** to the field and the strength of evidence to support the findings is **solid**.

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Abstract

Haemorrhagic fever with renal syndrome (HFRS) is a fatal disease caused by Hantaan virus (HTNV) infection. Humoral immunity is essential for effective viral clearance, however, the glycosylation characteristics of immunoglobulin G (IgG) in HFRS patients is not well known. Peripheral blood mononuclear cells from HFRS patients were obtained for B subset analysis using scRNA-seq and flow cytometry. HTNV specific IgG antibody titers were detected by enzyme-linked immunosorbent assay, and IgG glycosylation was analyzed by ultra-performance liquid chromatography. The proportions of the antibody-secreting memory (ASM) B cells and plasmablasts (PB) were significantly expanded among acute HFRS patients. We discovered significantly increased fucosylated IgG and decreased bisecting N-

acetylglucosamine during the convalescent phase of HTNV infection. Meanwhile, positive correlations were observed between ASM subsets and galactosylation/sialylation in the IgG Fc region, and between PB subsets and sialylation. Notably, the glycosylation-related genes, such as *RPN1* and *RPN2*, were primarily expressed differentially in the ASM and PB subclusters which were enriched in the N-glycosylation modifications of proteins through asparagine. Our findings indicated that IgG N-glycosylation may play a crucial role in combating HTNV infection and contributing to clinical recovery, which provided new insights for optimizing glycoengineered therapeutic antibodies.

Introduction

The causative agent of haemorrhagic fever with renal syndrome (HFRS) is Hantaan virus, which belongs to the family *Hantaviridae* and is primarily transmitted by rodents (Zheng et al., 2019). China is the most heavily affected country by HFRS, accounting for >90% of global cases, and a total of 32,462 cases of HFRS were reported in China during 2019–2022, with a fatality rate ranging from 0.1 to 15% (R. Li et al., 2023) (<https://www.chinacdc.cn>).

Hantaviruses are tri-segmented, single-stranded, negative-sense RNA viruses, whose genomes consist of three regions: large (L), medium (M), and small (S). The glycoproteins Gn and Gc, encoded by the M segment, can infect target cells - primarily vascular endothelial cells - via $\beta 3$ integrin receptors (Pizarro et al., 2019). Simultaneously, they could also infect other cell types, such as mononuclear macrophages and dendritic cells, leading to systemic viral infection. Although hantavirus replication is thought to occur primarily in the vascular endothelium without direct cytopathic effects, a plethora of innate immune cells mediate host antiviral defenses. These include natural killer cells, neutrophils, monocytes, and macrophages, together with pattern recognition receptors (PRRs), interferons (IFNs), antiviral proteins, and complement activation, e.g., via the pentraxin 3 (PTX3) pathway, which can exacerbate HFRS disease progression leading to immunopathological damage through cytokine/chemokine production, cytoskeletal rearrangements in endothelial cells, ultimately amplifying vascular dysfunction (Tariq & Kim, 2022). Rapid and effective humoral immune responses, however, such as neutralizing antibody responses targeting the glycoproteins Gn/Gc, contribute to rapid recovery from HFRS and are critical for protection from severe disease (Engdahl & Crowe, 2020; Li et al., 2020).

Immunoglobulin G (IgG) N-linked glycosylation mediates critical functions modulating antiviral immunity during viral infection. Changes in the conserved N-linked glycan Asn297 in the Fc region of IgG typically by fucosylation, galactosylation, or sialylation can alter antibody effector function. A reduction in core fucosylation decreases IgG binding to NK cell Fc γ RIIIa promotes antibody-dependent cellular cytotoxicity (ADCC) necessary for clearance of viruses, including SARS-CoV-2, dengue and HIV-1 whereas sialylation can attenuate immune responses resulting in immune evasion (Ash et al., 2022; Haslund-Gourley et al., 2024; Hou et al., 2021; Wang et al., 2017). Changes in IgG and other protein N-linked glycosylation profiles therefore shape virus-host interactions and disease progression.

Importantly, there have not been prior studies specifically examining plasma IgG N-glycome profiles derived from chromatographic peak data in HFRS patients, particularly in relation to seroconversion status. This gap in our knowledge motivated our systematic investigation of both total and virus-specific IgG glycosylation dynamics during acute infection. In the present study, we combined scRNA-seq and flow cytometry to reveal the phenotypes of B cell responses during Hantaan virus (HTNV) infection, and explored the transcriptomic features of reactive B cell subsets. A total of 166 HFRS patients, including 65 paired HFRS samples, were employed to profile the IgG-Fc glycosylation pattern and to investigate the potential regulatory role of IgG-Fc glycosylation in HFRS pathogenesis.

Results

B cell compositional characteristics in HFRS patients

To characterize the humoral immune profiles in HFRS patients, we enrolled 166 suspected HTNV-infected patients who were admitted to Baoji Central Hospital in Shaanxi Province, China, between October 2019 and January 2022. Among them, 65 met the inclusion criteria and were included in the study (**Figure 1**). We identified a total of eight cell subpopulations, including subsets of CD4 T cells, CD8 T cells, CD14 monocytes, NK cells, B cells, platelets, endothelial progenitor cells, and red blood cells (**Figure 2—figure supplement 1A** and **B**). The proportions of these cell types were comparable between acute HFRS patients and healthy controls, with no significant batch effects observed (**Figure 2—figure supplement 1C**). CD4 T cells (31.4%) were the predominant subset prior to HTNV infection, whereas CD8 T cells (44%) became the dominant population post-infection. Notably, the proportions of NK cells (15.9% to 5.6%, $p < 0.001$) and platelets (3.1% to 0.6%, $p < 0.01$) significantly decreased, whereas the level of CD14 monocytes (15.9% to 32.3%, $p < 0.05$) nearly doubled. The overall proportions of B cells, however, did not exhibit significant changes (**Figure 2—figure supplement 1D** and **E**).

To precisely elucidate the role of B cell subsets in anti-HTNV infection, we classified B cells into eight subsets, including antibody-secreting memory B cells (ASM), double-negative B cells (DN), intermediate memory B cells (IM), marginal zone-like B cells (MZB), naive B cells (naive B), plasmablasts (PB), quiescent resting memory B cells (RM), and exhausted tissue-like memory B cells (TLM) (**Figure 2A** and **B**, and **Figure 2—figure supplement 1F**). We discovered that significant expansion of ASM, PB, and RM cell populations in acute HFRS patients compared to healthy controls, accompanied by contraction of DN, IM, and naive B cell compartments post-infection (**Figure 2C**). In healthy controls, naive B cells (42.87%), DN B cells (26.65%), and TLM B cells (16.13%) constituted the majority of the B cell population. However, ASM (30.96%), PB cells (11.12%) and RM B cells (2.88%), became the dominant subsets after HTNV infection (**Figure 2D** and Supplementary file 1).

To validate the accuracy of our single-cell subpopulation analysis, we employed flow cytometry to quantify the B cell subsets in acute and convalescent HFRS patients (**Figure 2E**). The results demonstrated that ASM (39.48% vs. 24.57%, $p < 0.001$) and PB (23.97% vs. 2.61%, $p < 0.001$) subsets were significantly amplified in the acute phase compared to the convalescent phase (**Figure 2F**). Conversely, the number of naive B cells (32.78% vs. 48.92%, $p = 0.003$) and CSM cells (18.65% vs. 27.15%, $p = 0.008$) recovered rapidly in the convalescent stage (**Figure 2F**), consistent with the observations at the transcriptome level. No significant differences were observed for other B cell subtypes between the acute and convalescent stages (**Figure 2—figure supplement 2**). These results demonstrated that several B cell subpopulations were effectively activated following HTNV infection.

Dynamic characteristics of the IgG N-glycome in HFRS patients

HFRS patients were further stratified into three age groups (≤ 44 , 45–59, and ≥ 60 years). IgG glycosylation was analyzed by ultra-performance liquid chromatography. As shown in Supplementary file 2, no significant differences in IgG glycosylation were observed between age groups, except for galactosylation ($p = 0.001$). Similarly, no sex-based differences in IgG-Fc glycosylation were noted (Supplementary file 3). However, the levels of IgG with bisecting GlcNAc (15.72% vs. 14.51%, $p < 0.001$), galactosylated IgG (74.26% vs. 71.15%, $p < 0.001$), and sialylated IgG (22.44% vs. 21.62%, $p < 0.001$) were significantly higher in the acute phase compared to the convalescent phase. Conversely, fucosylated IgG levels (94.29% vs. 94.89%, $p < 0.001$) were lower in the acute phase (Supplementary file 4), indicating that IgG N-glycosylation plays a critical role in the anti-infection process. Multivariate linear regression was employed to mitigate potential

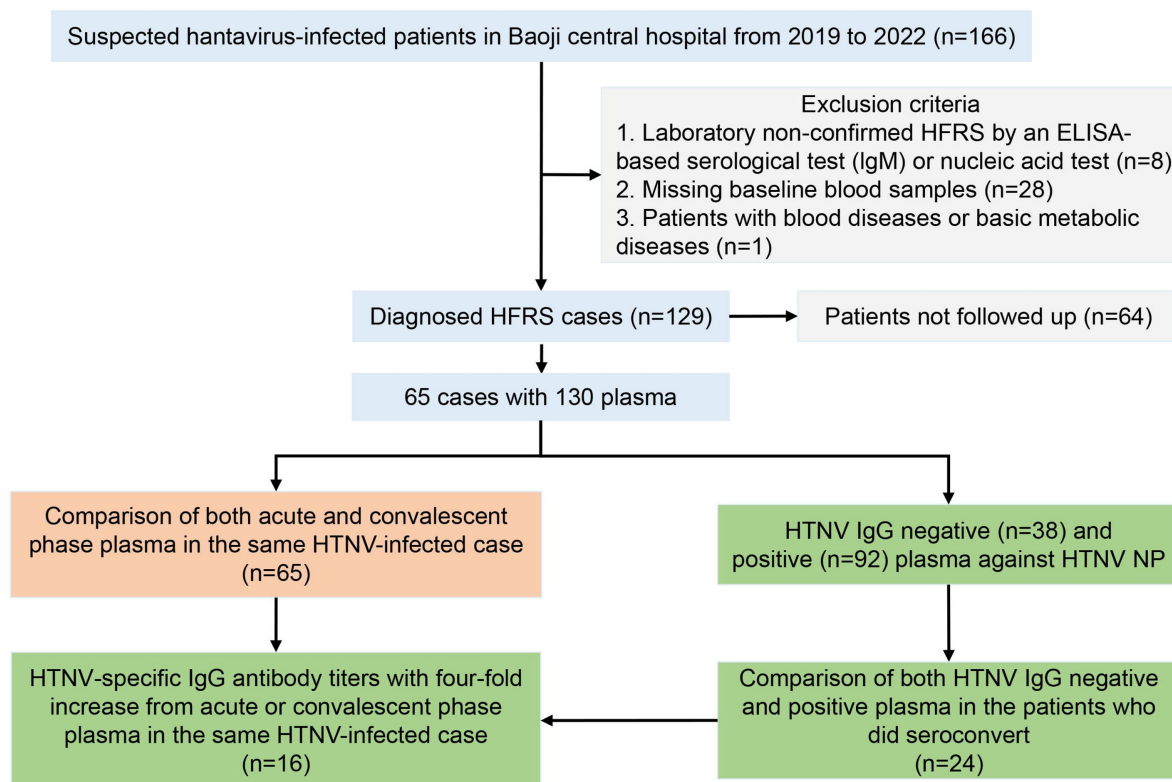


Figure 1.

Flow chart of HFRS patient recruitment in Baoji Central Hospital in Shaanxi province from November 2019 to January 2022.

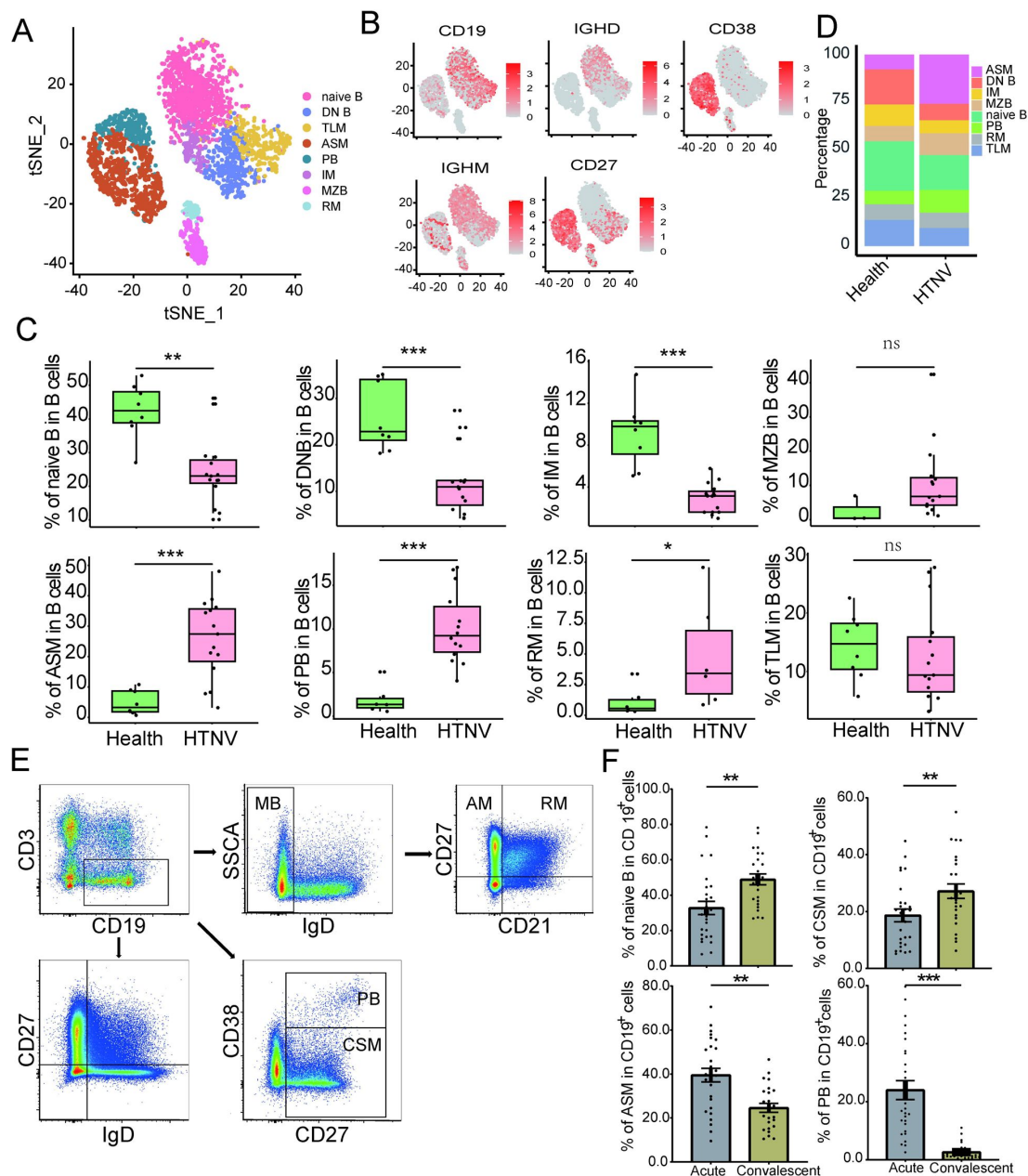


Figure 2.

Immunophenotypic remodeling in the B-cell subsets during acute HFERS.

(A) t-distributed stochastic neighbor embedding (t-SNE) plot showing antibody secreting memory B cells (ASM), double negative (DN) B cells, intermediate memory cells (IM), marginal zone-like cells (MZB), naive B cells, plasmablasts (PB), quiescent resting memory cells (RM), and exhausted tissue-like memory B cells (TLM) of PBMCs identified using an integrated and classification analysis. (B) t-SNE projection of canonical markers, including CD19, CD27, CD38, IgD, and IgM. (C) Proportions of the eight B cell subsets, colored by the healthy group (green) and HTNV groups (red). Boxplot features: minimum box, 25th percentile; center, median; maximum box, 75th percentile. (D) Frequency of the eight B cell subsets in between the healthy group (n = 8) and HTNV groups (n = 15). (E) FACS gating strategy for the measurement of the B-cell subsets: activated memory B cells (CD21⁺ CD27⁺, AM), RM B cells (CD21⁺ CD27⁺), IM B cells (CD21⁺ CD27⁺), TLM B cells (CD21⁺ CD27⁺), naive B cells (CD27⁺ IgD⁺), MZB B cells (CD27⁺ IgD⁺), ASM B cells (CD27⁺ IgD⁺), DN B cells (CD27⁺ IgD⁺), PB (CD38⁺ CD27⁺), and class-switched memory B cells (CD38⁺ CD27⁺, CSM), respectively. (F) The proportion of ASM, CSM, naive B, and PB cells in CD19⁺ B cells in the acute and convalescent HFERS patients. Data are presented as mean $\pm$ SEM in panels C and F. *p < 0.05 (Student's t-test).

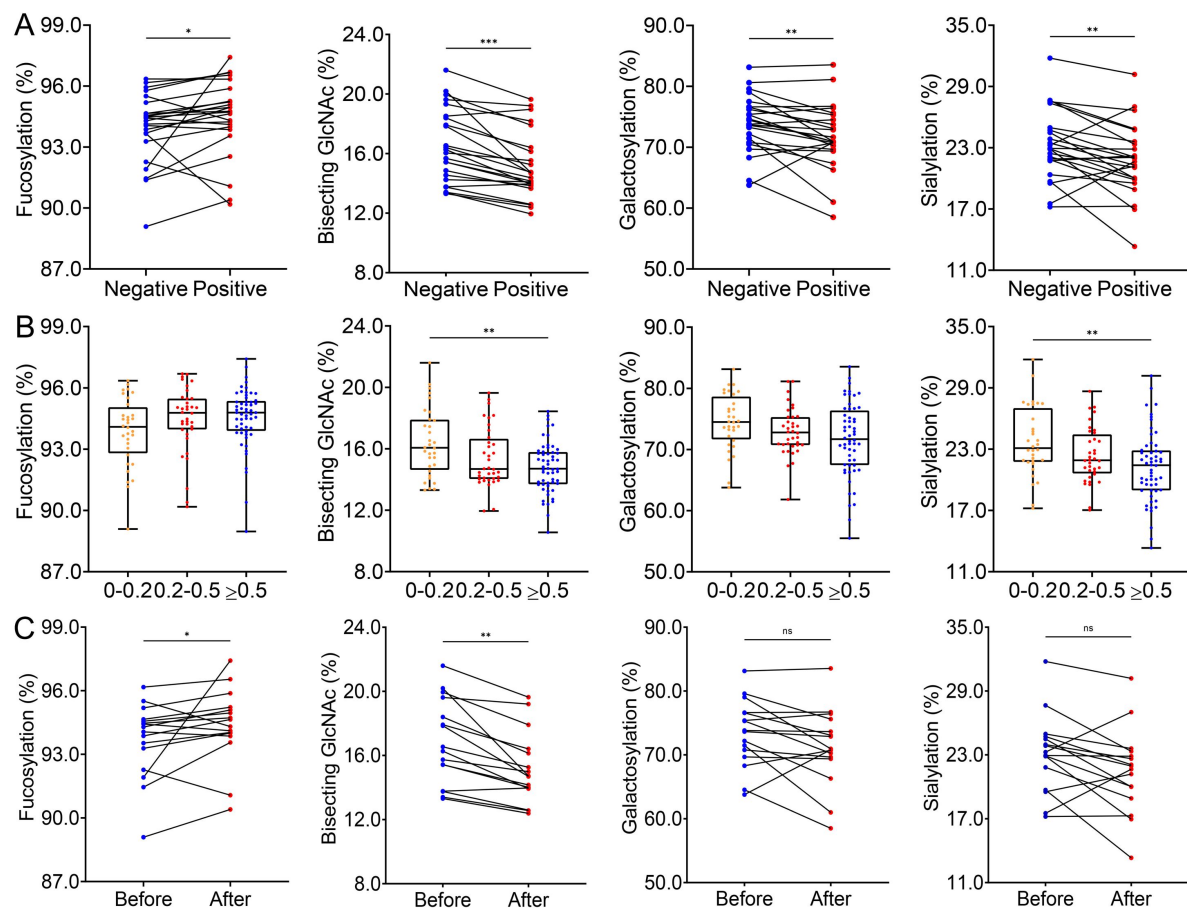


Figure 2—figure supplement 1.

Single-cell transcriptomes of PBMCs from patients with HFRS.

(A) Uniform manifold approximation and projection (UMAP) presentation of the major peripheral immune cell types in the PBMCs from 15 HFRS patients and 8 healthy controls. (B) Dot plot shows the expression levels of canonical marker-related genes in the different subpopulations. (C) UMAP presentation of the major peripheral immune cell types among HFRS and healthy control groups. (D) Proportions of the eight cell subsets, colored by the healthy group (green) and HTNV groups (red). Boxplot features: minimum box, 25th percentile; center, median; maximum box, 75th percentile. (E) Frequency of the eight cell subsets in between the healthy group ($n = 8$) and the HTNV group ($n = 15$). (F) The UMAP plot shows the differentiation trajectories of different cell types calculated by pseudotime analysis.

confounding by genetic and environmental factors in the glycomics analysis. While no significant associations were observed for most glycan models (fucosylation, $p=0.526$; bisecting GlcNAc, $p=0.069$; and sialylation, $p=0.058$), we discovered sex showed a potentially influential effect on galactosylation ($p=0.001$) (Supplementary files 5-8). These results suggest that while most glycan features appear unaffected by the examined covariates, galactosylation may be subject to sex-specific biological regulation.

Among the 130 paired blood samples from 65 patients, 38 (29.2%) were negative for HTNV nucleocapsid protein-specific IgG antibodies, and 92 (70.8%) were seropositive. For the 24 patients with both seronegative and seropositive samples, levels of bisecting GlcNAc (16.16% vs. 14.54%, $p<0.001$), galactosylation (74.12% vs. 70.94%, $p=0.006$), and sialylation (22.92% vs. 21.80%, $p=0.008$) were higher in the seronegative period, whereas that of fucosylation was lower (94.36% vs. 94.74%, $p=0.024$) in this period (**Figure 3A**). Notably, bisecting GlcNAc and sialylated IgG levels decreased as HTNV nucleocapsid protein (NP)-specific antibody titers increased (**Figure 3B**), highlighting a potential link between antibody levels and IgG-Fc glycosylation.

The IgG N-glycome in HFRS patients with seroconversion

A four-fold or greater increase in HTNV NP-specific antibody titers usually indicates a protective humoral immune response during the acute phase (Onwuchekwa et al., 2023). We observed a significant increase in the Fc fucosylation (94.18% vs. 94.48%, $p=0.044$) and a decrease in bisecting GlcNAc (16.41% vs. 14.71%, $p=0.001$) post the four-fold antibody titer increase compared to the baseline. However, no significant differences were observed in galactosylation and sialylation (**Figure 3C**), suggesting that fucosylation and bisecting GlcNAc may play key roles in anti-HTNV infection and recovery.

The potential source of the IgG N-glycome during HTNV infection

Positive correlations were observed between the ASM subsets and both galactosylation ($p=0.017$, $rs=0.418$) and sialylation ($p=0.008$, $rs=0.458$) in the antibody Fc region, as well as between the PB subsets and sialylation ($p=0.036$, $rs=0.372$) (**Figure 4A-C**). Therefore, we speculated that galactosylated antibodies may primarily originate from the ASM subsets, while sialylated antibodies may derive from the ASM and PB subsets.

To investigate the process of glycosylated antibody production secreted by the ASM and PB subsets, we identified differentially expressed genes (DEGs) among the eight B cell subpopulations. Notably, the most DEGs were observed in the ASM (470), naive (312), and TLM (333) B subpopulations, respectively (**Figure 4—figure supplement 1A** and Supplementary file 9). Specifically, among all the upregulated genes, the expressions of glycosylation-related, mitochondrial respiratory-related, and peptide chain elongation-related genes among the ASM, PB, and RM populations were elevated after HTNV infection, including *RPN2*, *MT-ND6*, and *RPS26* (**Figure 4—figure supplement 1B** and Supplementary file 10). On the contrary, the upregulated genes in the naive, DN, and IM populations were related to anti-inflammatory and immunosuppressive effects, inhibition of the antioxidative function, response to environmental stress, and B-cell growth and development, such as *TSC22D3*, *TXNIP*, *DUSP1*, and *KLF6*, whereas the downregulated genes played an important role in the inflammatory response, cell motility, and antigen presentation, including *LTB*, *ACTG1*, and *HLA-DRB5* (**Figure 4—figure supplement 1C** and Supplementary file 11).

Unexpectedly, we observed significant changes in the expression of glycosylation-related genes (GRGs) in the ASM, PB, and RM subclusters during HTNV infection compared to healthy controls. *RPN1* and *RPN2* were prominently upregulated, which encoded subunits of the oligosaccharyltransferase complex (**Figure 4D**). This implied that the process of glycosylation during HTNV infection was rapidly activated among the ASM and PB subclusters. In the ASM

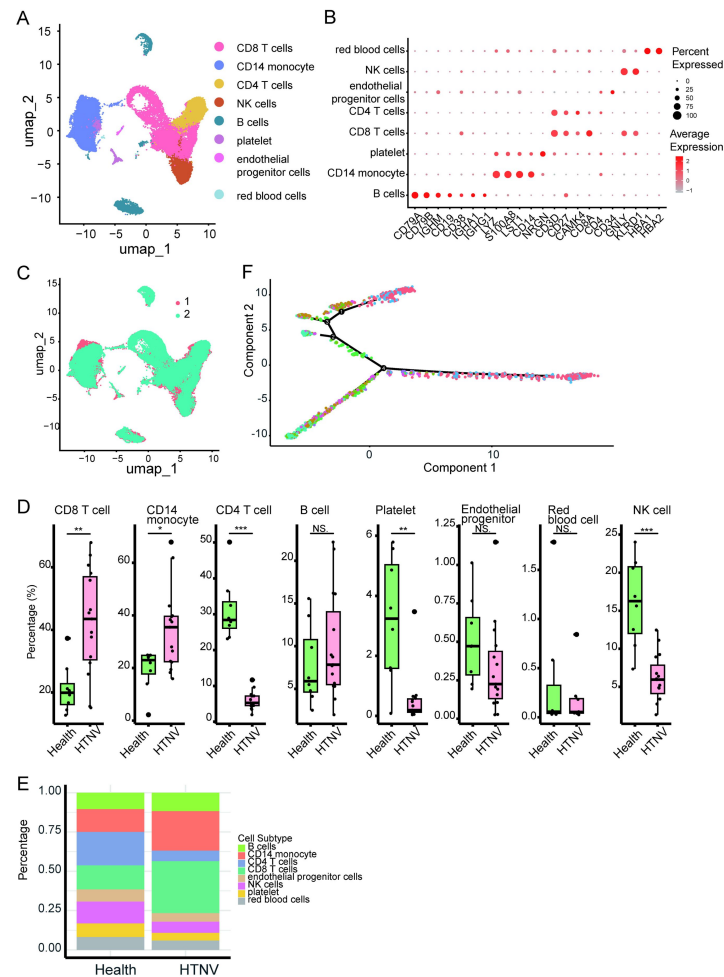


Figure 3.

Glycosylation modification of antibodies associated with HTNV infection.

(A) Changes of different glycosylation types in both HTNV NP-specific IgG negative and positive plasma from 24 HFRS patients. (B) Differential IgG glycosylation patterns across antibody titer levels quantified by ELISA. (C) Comparison of different glycosylation levels before and after the fourfold increase in the IgG antibody titers. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. For paired samples, Wilcoxon Signed Rank Test was used to assess the difference.

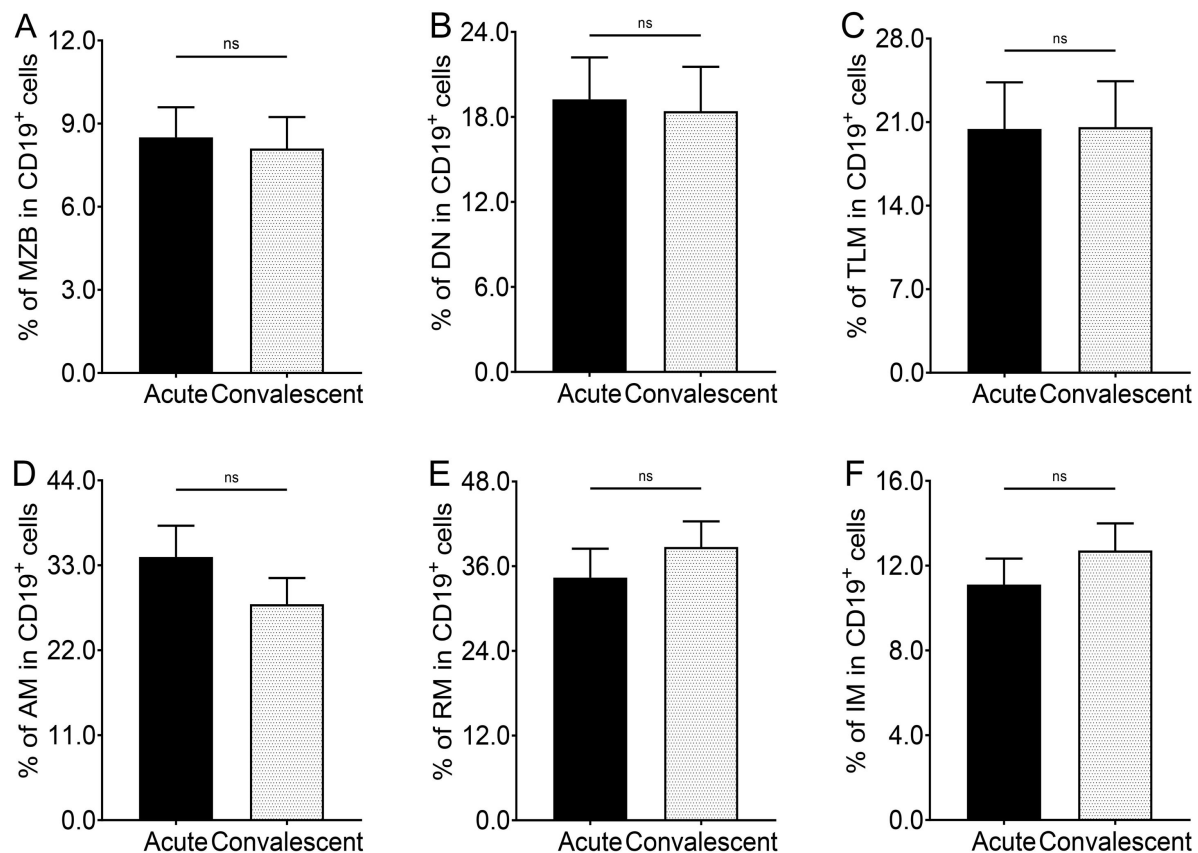


Figure 4.

Glycosylation modifications of antibodies primarily deprived from antibody-secreting and plasmablast subpopulations.

(A) The correlation between the proportion of PB cells and the sialylation level. (B) The correlation between the proportion of ASM cells and the sialylation level. (C) The correlation between the proportion of ASM cells and the galactosylation level. (D) Dot plot shows the expression levels of glycosylation-related genes in the eight B cell groups. The pink color represents mannose related genes, the green color represents N-glycosylation-related genes, the orange color represents galactosylation-related genes, the purple represents sialylation-related genes, and the red color represents fucosylation related genes. (E) Enriched pathways in the plasmablast and ASM subsets by GO enrichment analysis of the DEGs. (F) The GSEA map presents the enrichment score of GRGs in N-Glycan biosynthesis.

subcluster, the fold change of GRGs from all four glycosylation modification-related genes, aside from *RPN1* and *RPN2*, was similar, suggesting that ASM may undergo complex and diverse glycosylation modification processes.

To enhance our understanding of the transcriptomic characteristics of GRG-high-expressing PB and ASM subclusters, we conducted a comprehensive pathway enrichment analysis and identified significant enrichment of pathways related to glycosylation modifications, including the endoplasmic reticulum protein-containing complex, rough endoplasmic reticulum, and protein N-linked glycosylation via asparagine (**Figure 4E**). Additionally, gene sets associated with glycosylation modifications showed positive regulation of these pathways ($p=0.002$, **Figure 4F**). These results indicated that glycosylation modifications in the ASM, PB, and RM subpopulations were dynamically regulated during the acute phase of HFRS, likely to satisfy the amplified anti-HTNV antibody requirement and accelerate viral clearance.

Discussion

IgG glycosylation is a key regulator of inflammatory responses for infectious diseases, acting as a molecular switch between anti-inflammatory and pro-inflammatory effects upon antigenic challenge. One of the most significant findings of the present study is the elevation in IgG fucosylation and reduction in bisecting GlcNAc levels at the time of a four-fold increases in the HTNV NP-specific antibody titers. Afucosylation is known to enhance antibody-dependent cellular cytotoxicity (ADCC) by increasing the affinity of IgG-Fc for the FcγRIIIa receptor on NK cells, leading to heightened production of inflammatory cytokines such as IL-1β, IL-6, IFN-γ, and TNF-α (Larsen et al., 2021; Lu, Suscovich, Fortune, & Alter, 2018; Wieczorek, Braicu, Oliveira-Ferrer, Sehouli, & Blanchard, 2020). Increase in bisecting GlcNAc linked to Fc induces enhanced affinity for the FcγRIIIa receptor and enhanced ADCC function of the antibodies (Giron et al., 2020; Hou et al., 2021; Irvine & Alter, 2020). However, rather than directly influencing the Fc receptor binding, this modification enhances antibody functionality through afucosylation, which subsequently increases the binding affinity to the FcγRIIIa receptor (Ackerman et al., 2018; Jennewein & Alter, 2017). Our findings provide evidence that the dynamic alterations in glycosylation patterns, characterized by increased fucosylation and decreased bisecting GlcNAc levels during viral infection, are involved in disease pathogenesis and clinical recovery, suggesting their dual utility as both diagnostic biomarkers and promising therapeutic targets.

We also observed elevated galactosylation levels in the acute phase compared to the convalescent phase, a phenomenon also reported in SARS-CoV-2 infections but not in chronic diseases and cancers (Hou et al., 2021; Liu et al., 2018). Increased galactosylation may expand the distance between CH2 domains, exposing key amino acids for FcγRIIIa binding and enhancing ADCC functions (Houde, Peng, Berkowitz, & Engen, 2010). Simultaneously, the increase in the ASM subset, which positively correlated with galactosylation, may reflect polyclonal activation of memory B cells driven by pattern recognition receptors and cytokines, leading to the production of galactosylated antibodies. Similarly, sialylation levels were higher in the acute patients, potentially linked to the expansion of the ASM and PB subsets. In this study, the PB subset was significantly elevated in the acute phase and positively correlated with sialylation, suggesting that rapidly proliferating plasmablasts may contribute to the secretion of sialylated antibodies; however, the mechanisms behind this require further investigation.

Antibody-related glycogenes are significantly activated following Hantaan virus infection. We noted that ribophorin I and II (*RPN1* and *RPN2*) were significantly upregulated in the ASM/IM/PB/RM subsets after Hantaan virus infection, which linked the high mannose oligosaccharides with asparagine residues found in the Asn-X-Ser/Thr consensus motif (Hwang et al., 2025). We speculate that they continuously attach the synthesized glycan chains to the constant region of antibodies during antibody synthesis. Similarly, fucosyltransferase 8 (*FUT8*) in

the ASM subset, catalyzing the alpha1-2, alpha1-3, and alpha1-4 fucose addition (Wang & Ravetch, 2019 [↗](#); Yang et al., 2015 [↗](#)), was downregulated in the mRNA translation, and the levels of fucosylated antibodies were naturally lower in the acute HFRS patients. Meanwhile, the beta-1,4-galactosyltransferase (beta4GalT) gene expression was significantly elevated in the ASM subpopulation during the acute phase, which also correlated with increased levels of galactosylated antibodies in serum (Wang & Ravetch, 2019 [↗](#)). However, we did not observe significant upward changes in sialyltransferase mRNA expression in the acute HFRS patients, similar with the finding from severe COVID-19 cohorts (Haslund-Gourley et al., 2024 [↗](#)). The neuraminidase 1 (NEU1) gene is strikingly upregulated and may potentially explain the decreased sialylation on the secreted HTNV-specific IgG antibodies during convalescence. Overall, the glycosylation of immunoglobulin G is regulated by a large network of B-cell glycogenes during HTNV infection.

Despite establishing a well-characterized patient cohort and performing systematic IgG glycosylation profiling based on HTNV NP antibody status, this study has several noteworthy limitations. Most notably, while preliminary comparisons suggested similar patterns between virus-specific and total IgG N-glycome, our total plasma IgG analysis may have introduced confounding factors in the observed associations. This methodological constraint could potentially affect the interpretation of certain disease-specific glycosylation signatures.

Our findings demonstrate that the IgG-Fc glycosylation plays a pivotal role in HTNV infection and rapid recovery. Fucosylated oligosaccharides and the absence of bisecting GlcNAc are associated with favorable clinical outcomes. Importantly, specific B cell subsets, particularly ASMs and PBs, upregulated the secretion of galactosylated and sialylated antibodies, respectively. These findings would deepen our understanding of antibody glycosylation in antiviral immunity and provide a foundation for future studies aimed at optimizing glycoengineered therapeutic antibodies.

Materials and Methods

Study participants

Clinical specimens were collected from HFRS patients who were hospitalized in Baoji Central Hospital between October 2019 and January 2022. Patients were categorized into four clinical subtypes (mild, moderate, severe, and critical) based on the diagnostic criteria for HFRS issued by the Ministry of Health (Ma et al., 2015 [↗](#)). This study was approved by the ethics committee of the Shandong First Medical University & Shandong Academy of Medical Sciences (R201937). Written informed consent was obtained from each participant or their guardians.

The clinical course of HFRS is grouped into acute (febrile, hypotensive, and oliguric stages) and convalescent (diuretic and convalescent stages) phases. The acute phase was defined as within 12 days of illness onset, and the convalescent phase was defined as a period of illness lasting 13 days or longer (Tang et al., 2019 [↗](#); Zhang et al., 2022 [↗](#)). The earliest sample was selected if there were multiple blood samples available in the acute phase and the last available sample before discharge was selected if there were multiple blood samples in the convalescent phase.

Single-Cell RNA Sequencing and public database

Flow cytometry antibody staining was conducted as previously described (Chakraborty et al., 2022 [↗](#)). Dead cell and red blood cell were removed using the BD FACSAria™ III cell sorter, and peripheral blood mononuclear cells (PBMCs) were collected for single-cell sequencing. The library construction was conducted as previously described (Jin et al., 2024 [↗](#)).

We obtained single-cell transcriptome sequencing data from the Gene Expression Omnibus (GEO) database (GSE161354), containing six HFRS patients and two healthy volunteers, as well as healthy control samples from the Genome Sequence Archive (GSA) database (HRA000203), matched with patient demographic information.

Single-Cell Data Analysis

The FastQC software was used to evaluate the data obtained to ensure the quality of the raw sequencing data. The raw data were mapped to the human reference genome (GRCh38, <https://cf.10xgenomics.com/supp/cell-exp/refdata-gex-GRCh38-2020-A.tar.gz>) using Cell Ranger. The Seurat R package (Version 4.4.0) was utilized for the merging and clustering of single-cell data. The Harmony R package and the anchor module of Seurat were used to remove batch effects between samples and groups for cell clustering. t-distributed Stochastic Neighbor Embedding (t-SNE) and Uniform Manifold Approximation and Projection (UMAP) were employed for dimensionality reduction and visualization of individual cells.

Cell Type Annotation

UMAP and t-SNE were used to reduce the dimension of all cells and to cluster them in a two-dimensional space based on shared features. The Single R (Version 2.2.0) package was used to infer independently the cell origin of each single cell based on a reference transcriptome dataset of pure cell types, for unbiased cell type identification from single-cell RNA sequencing data. Subsequently, specific highly-expressed genes were combined with those derived from review of the literature for manual annotation to ultimately determine cell types. Classic biomarkers for specific cell types were used to identify cells in different clusters.

Differential Gene Identification and Functional Analysis

The FindMarkers() function in the Seurat package was used to identify DEGs between different cell groups, with criteria of $|\log_2FC| > 0.25$ and $p\text{-value} < 0.05$, $pct.1 > 0.25$. The ClusterProfiler R package was utilized for GO, KEGG, and GSEA enrichment analyses, and the ggplot2 and enrichplot R packages were used for visualization.

Measurement of HTNV specific antibodies

Levels of serum IgM and IgG antibodies to HTNV NP were assessed through the enzyme-linked immunosorbent assay (ELISA) kit (Wantai BioPharm, Beijing, China) in accordance with the manufacturer's instructions.

Analysis of IgG glycans

The process of isolating, labelling, purifying, and analyzing plasma IgG was performed using previously established methods (Hou et al., 2019; Pengcheng Liu, 2023). Briefly, the diluted plasma samples were transferred onto a 96-well protein G monolithic plate (BIA Separations, Slovenia) for the isolation of IgG. The isolated IgG was eluted with 1 mL of 0.1 M formic acid and was immediately neutralized with 170 μ L of 1M ammonium bicarbonate.

The released N-glycans were labelled with 2-aminobenzamide (2-AB) and were then purified from a mixture of 100% acetonitrile and ultrapure water in a 1:1 ratio (v/v). This was then analyzed by hydrophilic interaction liquid chromatography using ultra-performance liquid chromatography (HILIC-UPLC; Waters Corporation, Milford, MA) (Hou et al., 2019). As previously reported, the chromatograms were separated into 24 IgG glycan peaks (GPs) (Menni et al., 2018).

Our study incorporated both biological and technical replicates to ensure a robust glycomic profiling analysis. Specifically, we analyzed paired acute/convalescent-phase samples from 65 confirmed HFRS patients to assess inter-individual biological variability, while technical

reproducibility was validated through comparison with standard chromatographic peak plots (Vučković et al., 2016 [↗](#)). This dual-replicate strategy enabled a comprehensive evaluation of both biological heterogeneity and assay precision.

Virus-specific B-cell detection

Virus-specific B-cell against HTNV were performed as previously described (Liechti & Roederer, 2019 [↗](#); Song et al., 2018 [↗](#)).

Statistical analysis

Categorical variables were compared using chi-square tests. Normality of numeric variables was assessed with the Kolmogorov-Smirnov test; non-normally distributed continuous variables were analyzed using non-parametric tests. The Benjamini - Hochberg (BH) method was used to adjust the raw p-values from DEG analysis, controlling the false discovery rate (FDR). Statistical analyses were performed with R software version 4.1.1 (R Core Team, New Zealand) and GraphPad Prism 8 (GraphPad Software, San Diego, CA), with a significance level of $\alpha = 0.05$ (two-tailed).

Data availability

The data underlying this article will be shared upon request to the corresponding author.

Additional information

Author contributions

Haifeng Hou and Weifeng Shi designed and supervised this study. Chuansong Quan, Lu Wang, Jiming Gao, Yaoni Li, Houqiang Li, Hongzhi Liu, Qian Li, and Liqiong Zhao collected the samples and performed experiments. Chuansong Quan, Lu Wang, Jiming Gao, Xiaoyu Xu, Zixuan Gao, Wenxu Ruan, and Weijia Xing analyzed the data and performed the statistical analyses. Lu Wang, Chuansong Quan, Jiming Gao, and Xiaoyu Xu wrote the manuscript. Haifeng Hou, Weifeng Shi, Weijia Xing, and Michael J. Carr reviewed and edited the manuscript. All authors had full access to all of the data in the study and accept responsibility for the decision to submit for publication.

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Additional files

Supplementary file 1. The proportion of different B cell subpopulations between healthy and acute HFRS patients. [↗](#)

Supplementary file 2. Difference in main IgG glycome features of HFRS patients between age groups. [↗](#)

Supplementary file 3. Difference in main IgG glycome features of HFRS patients between sex groups. [↗](#)

Supplementary file 4. Relative abundance (%) of the main IgG-Fc glycome features in HFRS patients. [↗](#)

Supplementary file 5. Results of multiple linear regression analysis for fucosylation. APTT, activated partial thromboplastin time; MCHC, mean corpuscular hemoglobin concentration; NEUT, neutrophil; PLT, platelet; WBC, white blood cell; CRE, creatinine; Cys-C, cystatin C. [↗](#)

Supplementary file 6. Results of multiple linear regression analysis for bisecting GlcNAc. APTT, activated partial thromboplastin time; MCHC, mean corpuscular hemoglobin concentration; NEUT, neutrophil; PLT, platelet; WBC, white blood cell; CRE, creatinine; Cys-C, cystatin C. [↗](#)

Supplementary file 7. Results of multiple linear regression analysis for galactosylation. APTT, activated partial thromboplastin time; MCHC, mean corpuscular hemoglobin concentration; NEUT, neutrophil; PLT, platelet; WBC, white blood cell; CRE, creatinine; Cys-C, cystatin C. [↗](#)

Supplementary file 8. Results of multiple linear regression analysis for sialylation. APTT, activated partial thromboplastin time; MCHC, mean corpuscular hemoglobin concentration; NEUT, neutrophil; PLT, platelet; WBC, white blood cell; CRE, creatinine; Cys-C, cystatin C. [↗](#)

Supplementary file 9. Differentially expressed genes of different B cell subgroups a between healthy and acute HFRS patients. [↗](#)

Supplementary file 10. Differential expression genes in the ASM, PB and RM B cell subpopulations. [↗](#)

Supplementary file 11. Differential expression genes in the DN, IM and naive B cell subpopulations. [↗](#)

References

1. Zheng Z., Wang P., Wang Z., Zhang D., Wang X., Zuo S., Li X (2019) **The characteristics of current natural foci of hemorrhagic fever with renal syndrome in Shandong Province, China, 2012-2015** *PLoS Negl Trop Dis* **13**:e0007148 <https://doi.org/10.1371/journal.pntd.0007148> | [Google Scholar](#)
2. Li R., Sun J., Chen Y., Fan X., Wang X., Zhang X., ..., Liu Z. (2023) **Clinical and laboratory features and factors predicting disease severity in pediatric patients with hemorrhagic fever with renal syndrome caused by Hantaan virus** *J Med Virol* **95**:e28339 <https://doi.org/10.1002/jmv.28339> | [Google Scholar](#)
3. Pizarro E., Navarrete M., Mendez C., Zaror L., Mansilla C., Tapia M., ..., Rodríguez E. M. (2019) **Immunocytochemical and ultrastructural evidence supporting that Andes Hantavirus (ANDV) is transmitted person-to-person through the respiratory and/or salivary pathways** *Front Microbiol* **10**:2992 <https://doi.org/10.3389/fmicb.2019.02992> | [Google Scholar](#)
4. Tariq M., Kim D. M (2022) **Hemorrhagic fever with renal syndrome: literature review, epidemiology, clinical picture and pathogenesis** *Infect Chemother* **54**:1–19 <https://doi.org/10.3947/ic.2021.0148> | [Google Scholar](#)
5. Engdahl T. B., Crowe J. E. (2020) **Humoral immunity to Hantavirus infection** *mSphere* **5** <https://doi.org/10.1128/mSphere.00482-20> | [Google Scholar](#)

6. Li Y., Quan C., Xing W., Wang P., Gao J., Zhang Z., ..., Shi W. (2020) **Rapid humoral immune responses are required for recovery from haemorrhagic fever with renal syndrome patients** *Emerg Microbes Infect* **9**:2303–2314 <https://doi.org/10.1080/22221751.2020.1830717> | Google Scholar
7. Ash M. K., Bhimalli P. P., Cho B. K., Mattamana B. B., Gambut S., Tarhoni I., ..., Schneider J. R. (2022) **Bulk IgG glycosylation predicts COVID-19 severity and vaccine antibody response** *Cell Rep* **41**:111799 <https://doi.org/10.1016/j.celrep.2022.111799> | Google Scholar
8. Haslund-Gourley B. S., Woloszczuk K., Hou J., Connors J., Cusimano G., Bell M., ..., Comunale M. A. (2024) **IgM N-glycosylation correlates with COVID-19 severity and rate of complement deposition** *Nat Commun* **15**:404 <https://doi.org/10.1038/s41467-023-44211-0> | Google Scholar
9. Hou H., Yang H., Liu P., Huang C., Wang M., Li Y., ..., Wang W. (2021) **Profile of immunoglobulin G N-glycome in COVID-19 patients: a case-control study** *Front Immunol* **12**:748566 <https://doi.org/10.3389/fimmu.2021.748566> | Google Scholar
10. Wang T. T., Sewatanon J., Memoli M. J., Wrammert J., Bournazos S., Bhaumik S. K., ..., Ravetch J. V. (2017) **IgG antibodies to dengue enhanced for FcγRIIIA binding determine disease severity** *Science* **355**:395–398 <https://doi.org/10.1126/science.aai8128> | Google Scholar
11. Onwuchekwa C., Moreo L. M., Menon S., Machado B., Curcio D., Kalina W., ..., Begier E. (2023) **Underascertainment of respiratory syncytial virus infection in adults due to diagnostic testing limitations: a systematic literature review and meta-analysis** *J Infect Dis* **228**:173–184 <https://doi.org/10.1093/infdis/jiad012> | Google Scholar
12. Larsen M. D., de Graaf E. L., Sonneveld M. E., Plomp H. R., Nouta J., Hoepel W., ..., Vidarsson G. (2021) **Afucosylated IgG characterizes enveloped viral responses and correlates with COVID-19 severity** *Science* **371** <https://doi.org/10.1126/science.abc8378> | Google Scholar
13. Lu L. L., Suscovich T. J., Fortune S. M., Alter G (2018) **Beyond binding: antibody effector functions in infectious diseases** *Nat Rev Immunol* **18**:46–61 <https://doi.org/10.1038/nri.2017.106> | Google Scholar
14. Wieczorek M., Braicu E. I., Oliveira-Ferrer L., Sehouli J., Blanchard V (2020) **Immunoglobulin G subclass-specific glycosylation changes in primary epithelial ovarian cancer** *Front Immunol* **11**:654 <https://doi.org/10.3389/fimmu.2020.00654> | Google Scholar
15. Giron L. B., Colomb F., Papasavvas E., Azzoni L., Yin X., Fair M., ..., Abdel-Mohsen M. (2020) **Interferon-α alters host glycosylation machinery during treated HIV infection** *EBioMedicine* **59**:102945 <https://doi.org/10.1016/j.ebiom.2020.102945> | Google Scholar
16. Irvine E. B., Alter G (2020) **Understanding the role of antibody glycosylation through the lens of severe viral and bacterial diseases** *Glycobiology* **30**:241–253 <https://doi.org/10.1093/glycob/cwaa018> | Google Scholar
17. Ackerman M. E., Das J., Pittala S., Broge T., Linde C., Suscovich T. J., ..., Alter G. (2018) **Route of immunization defines multiple mechanisms of vaccine-mediated protection against SIV** *Nat Med* **24**:1590–1598 <https://doi.org/10.1038/s41591-018-0161-0> | Google Scholar
18. Jennewein M. F., Alter G (2017) **The immunoregulatory roles of antibody glycosylation** *Trends Immunol* **38**:358–372 <https://doi.org/10.1016/j.it.2017.02.004> | Google Scholar

19. Liu D., Chu X., Wang H., Dong J., Ge S. Q., Zhao Z. Y., ..., Wang W. (2018) **The changes of immunoglobulin G N-glycosylation in blood lipids and dyslipidaemia** *J Transl Med* **16**:235 <https://doi.org/10.1186/s12967-018-1616-2> | Google Scholar
20. Houde D., Peng Y., Berkowitz S. A., Engen J. R (2010) **Post-translational modifications differentially affect IgG1 conformation and receptor binding** *Mol Cell Proteomics* **9**:1716–1728 <https://doi.org/10.1074/mcp.M900540-MCP200> | Google Scholar
21. Hwang H. J., Kang D., Shin J., Jung J., Ko S., Jung K. H., ..., Lee J. S. (2025) **Therapy-induced senescent cancer cells contribute to cancer progression by promoting ribophorin 1-dependent PD-L1 upregulation** *Nat Commun* **16**:353 <https://doi.org/10.1038/s41467-024-54132-1> | Google Scholar
22. Wang T. T., Ravetch J. V (2019) **Functional diversification of IgGs through Fc glycosylation** *J Clin Invest* **129**:3492–3498 <https://doi.org/10.1172/jci130029> | Google Scholar
23. Yang Z., Wang S., Halim A., Schulz M. A., Frodin M., Rahman S. H., ..., Clausen H. (2015) **Engineered CHO cells for production of diverse, homogeneous glycoproteins** *Nat Biotechnol* **33**:842–844 <https://doi.org/10.1038/nbt.3280> | Google Scholar
24. Ma Y., Yuan B., Zhuang R., Zhang Y., Liu B., Zhang C., ..., Jin B. (2015) **Hantaan virus infection induces both Th1 and ThGranzyme B+ cell immune responses that associated with viral control and clinical outcome in humans** *PLoS Pathog* **11**:e1004788 <https://doi.org/10.1371/journal.ppat.1004788> | Google Scholar
25. Tang K., Zhang C., Zhang Y., Zhang Y., Du H., Jin B., Ma Y (2019) **Elevated plasma interleukin 34 levels correlate with disease severity-reflecting parameters of patients with haemorrhagic fever with renal syndrome** *Infect Dis (Lond)* **51**:847–853 <https://doi.org/10.1080/23744235.2019.1672887> | Google Scholar
26. Zhang H., Wang Y., Ma Y., Tang K., Zhang C., Wang M., ..., Zhang Y. (2022) **Increased CD4(+)CD8(+) double positive T cells during Hantaan virus infection** *Viruses* **14** <https://doi.org/10.3390/v14102243> | Google Scholar
27. Chakraborty S., Gonzalez J. C., Sievers B. L., Mallajosyula V., Chakraborty S., Dubey M., ..., Wang T. T. (2022) **Early non-neutralizing, afucosylated antibody responses are associated with COVID-19 severity** *Sci Transl Med* **14**:eabm7853 <https://doi.org/10.1126/scitranslmed.abm7853> | Google Scholar
28. Jin C., Chen Y., Wang Y., Li J., Liang J., Zheng S., ..., Yang H. (2024) **Single-cell RNA sequencing reveals special basal cells and fibroblasts in idiopathic pulmonary fibrosis** *Sci Rep* **14**:15778 <https://doi.org/10.1038/s41598-024-66947-5> | Google Scholar
29. Hou H., Xu X., Sun F., Zhang X., Dong H., Wang L., ..., Wang Y. (2019) **Hyperuricemia is associated with immunoglobulin G N-glycosylation: a community-based study of glycan biomarkers** *Omic* **23**:660–667 <https://doi.org/10.1089/omi.2019.0004> | Google Scholar
30. Liu Pengcheng, Aishe Dun X. W., Li Yutong, Li Houqiang, Wang Lu, Zhang Yichun, Li Cancan, Zhang Jinxia, Zhang Xiaoyu, Ma f Lixing, Hou Haifeng (2023) **High-throughput profiling of serological immunoglobulin G N-glycome as a noninvasive biomarker of gastrointestinal cancers** *Engineering* **26**:44–53 <https://doi.org/10.1016/j.eng.2023.02.008> | Google Scholar
31. Menni C., Gudelj I., Macdonald-Dunlop E., Mangino M., Zierer J., Bešić E., ..., Valdes A. M. (2018) **Glycosylation profile of immunoglobulin G is cross-sectionally associated with**

cardiovascular disease risk score and subclinical atherosclerosis in two independent cohorts *Circ Res* **122**:1555–1564 <https://doi.org/10.1161/circresaha.117.312174> | Google Scholar

32. Vučković F., Theodoratou E., Thaçi K., Timofeeva M., Vojta A., Štambuk J., ..., Lauc G. (2016) **IgG glycome in colorectal cancer** *Clin Cancer Res* **22**:3078–3086 <https://doi.org/10.1158/1078-0432.Ccr-15-1867> | Google Scholar
33. Liechti T., Roederer M (2019) **OMIP-051 - 28-color flow cytometry panel to characterize B cells and myeloid cells** *Cytometry A* **95**:150–155 <https://doi.org/10.1002/cyto.a.23689> | Google Scholar
34. Song P., Zheng N., Liu Y., Tian C., Wu X., Ma X., ..., Wu Z. (2018) **Deficient humoral responses and disrupted B-cell immunity are associated with fatal SFTSV infection** *Nat Commun* **9**:3328 <https://doi.org/10.1038/s41467-018-05746-9> | Google Scholar

Zhang Y, Tang K (2021) **A single-cell atlas of the peripheral immune response in patients with HFRS** *NCBI Gene Expression Omnibus*

Su Wenru (2020) **A human immune cell landscape in aging and COVID-19** *CNCB Genome Sequence Archive*

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Reviewer #1 (Public review):

Summary:

The authors investigated the potential role of IgG N-glycosylation in Haemorrhagic Fever with Renal Syndrome (HFRS), which may offer significant insights for understanding molecular mechanisms and for the development of therapeutic strategies for this infectious disease.

Comments on revisions:

While the majority of the issues have been addressed, a few minor points still remain unresolved.

Quality control should be conducted prior to the analysis of clinical samples. However, the coefficient of variation (CV) value was not provided for the paired acute and convalescent-phase samples from 65 confirmed HFRS patients, which were analyzed to assess inter-individual biological variability. It is important to note that biological replication should be evaluated using general samples, such as standard serum.

<https://doi.org/10.7554/eLife.106989.2.sa2>

Reviewer #2 (Public review):

This work sought to explore antibody responses in the context of hemorrhagic fever with renal syndrome (HFRS) - a severe disease caused by Hantaan virus infection. Little is known about the characteristics or functional relevance of IgG Fc glycosylation in HFRS. To address this gap, the authors analyzed samples from 65 patients with HFRS spanning the acute and convalescent phases of disease via IgG Fc glycan analysis, scRNAseq, and flow cytometry. The authors observed changes in Fc glycosylation (increased fucosylation and decreased bisection) coinciding with a 4-fold or greater increase in Hantaan virus-specific antibody titer. The study also includes exploratory analyses linking IgG glycan profiles to glycosylation-related gene expression in distinct B cell subsets, using single-cell transcriptomics. Overall, this is an interesting study that combines serological profiling with transcriptomic data to shed light on humoral immune responses in an underexplored infectious disease. The integration of Fc glycosylation data with single-cell transcriptomic data is a strength.

The authors have addressed the major concerns from the initial review. However, one point to emphasize is that the data are correlative. While the associations between Fc glycosylation changes and recovery are intriguing, the evidence does not establish causation. This is not a

weakness, as correlative studies can still be highly valuable and informative. However, the manuscript would be strengthened by making this distinction clear, particularly in the title.

<https://doi.org/10.7554/eLife.106989.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public review):

(1) The authors should provide a detailed description of the pathogenesis of Haemorrhagic Fever with Renal Syndrome (HFRS) and elaborate on the crucial role of IgG proteins in the disease's progression (line 65).

As suggested, we have now provided a detailed description of the pathogenesis of HFRS and elaborated on the crucial role of IgG proteins in the disease's progression:

"Hantaviruses are tri-segmented, single-stranded, negative-sense RNA viruses, whose genomes consist of three regions: large (L), medium (M), and small (S). The glycoproteins Gn and Gc, encoded by the M segment, can infect target cells - primarily vascular endothelial cells - via $\beta 3$ integrin receptors (Pizarro et al., 2019). Simultaneously, they could also infect other cell types, such as mononuclear macrophages and dendritic cells, leading to systemic viral infection. Although hantavirus replication is thought to occur primarily in the vascular endothelium without direct cytopathic effects, a plethora of innate immune cells mediate host antiviral defenses. These include natural killer cells, neutrophils, monocytes, and macrophages, together with pattern recognition receptors (PRRs), interferons (IFNs), antiviral proteins, and complement activation, e.g., via the pentraxin 3 (PTX3) pathway, which can exacerbate HFRS disease progression leading to immunopathological damage through cytokine/chemokine production, cytoskeletal rearrangements in endothelial cells, ultimately amplifying vascular dysfunction (Tariq & Kim, 2022). Rapid and effective humoral immune responses, however, such as neutralizing antibody responses targeting the glycoproteins Gn/Gc, contribute to rapid recovery from HFRS and are critical for protection from severe disease (Engdahl & Crowe, 2020; Li et al., 2020)." Please see the Introduction (Page 4, lines 65-81).

(2) An additional discussion on the significance of glycosylation, particularly IgG N-glycosylation, in viral infections should be included in the Introduction section.

Thank you for the suggestion and we have added an additional discussion on the significance of glycosylation in viral infections in the revised Introduction section.

"Immunoglobulin G (IgG) N-linked glycosylation mediates critical functions modulating antiviral immunity during viral infection. Changes in the conserved N-linked glycan Asn297 in the Fc region of IgG typically by fucosylation, galactosylation, or sialylation can alter antibody effector function. A reduction in core fucosylation decreases IgG binding to NK cell Fc γ RIIIa promotes antibody-dependent cellular cytotoxicity (ADCC) necessary for clearance of viruses, including SARS-CoV-2, dengue and HIV-1 whereas sialylation can attenuate immune responses resulting in immune evasion (Ash et al., 2022; Haslund-Gourley et al., 2024; Hou et al., 2021; Wang et al., 2017). Changes in IgG and other protein N-linked glycosylation profiles therefore shape virus-host interactions and disease progression." (Page 4, lines 82-91).

(3) In the abstract section, the authors state that HTNV-specific IgG antibody titers were detected and IgG N-glycosylation was analyzed. However, the analysis of plasma IgG N-glycans is described in the Methods section. Therefore, the authors should clarify the

glycome analysis process. Was the specific IgG glycome profile similar to the total IgG N-glycome? Given the biological relevance of specific IgG in immunological diseases, characterizing the specific IgG N-glycome profile would be more significant than analyzing the total plasma IgG.

We are grateful to the reviewer for the comments. Previous studies on viral infections have revealed that the pattern of virus-specific IgG N-glycans may be similar to that of total IgG N-glycome, and we therefore analyzed the total plasma IgG glycosylation profiling in the HFRS patients. However, we have discussed this in the Discussion section.

"Despite establishing a well-characterized patient cohort and performing systematic IgG glycosylation profiling based on HTNV NP antibody status, this study has several noteworthy limitations. Most notably, while preliminary comparisons suggested similar patterns between virus-specific and total IgG N-glycome, our total plasma IgG analysis may have introduced confounding factors in the observed associations. This methodological constraint could potentially affect the interpretation of certain disease-specific glycosylation signatures." Please see the Discussion (Page 12, lines 274-280).

References

- (1) Mads Delbo Larsen, Erik L de Graaf, Myrthe E Sonneveld, et al. Afucosylated IgG characterizes enveloped viral responses and correlates with COVID-19 severity. *Science*. 2021 Feb 26;371(6532):eabc8378.
- (2) Chakraborty S, Gonzalez J, Edwards K, et al. Proinflammatory IgG Fc structures in patients with severe COVID-19. *Nat Immunol*. 2021 Jan;22(1):67-73.
- (3) Tea Petrović, Amrita Vijay, Frano Vučković, et al. IgG N-glycome changes during the course of severe COVID-19: An observational study. *EBioMedicine*. 2022 Jul ;81: 104101.
- (4) Hou H, Yang H, Liu P, et al. Profile of Immunoglobulin G N-Glycome in COVID-19 Patients: A Case-Control Study. *Front Immunol*. 2021 Sep 23;12:748566.

(4) Further details regarding the N-glycome analysis should be provided, including the quantity of IgG protein used and the methodology employed for analyzing IgG N-glycans (lines 286-287).

As suggested, we have provided further details regarding the N-glycome analysis in the Method section.

"Briefly, the diluted plasma samples were transferred onto a 96-well protein G monolithic plate (BIA Separations, Slovenia) for the isolation of IgG. The isolated IgG was eluted with 1 mL of 0.1 M formic acid and was immediately neutralized with 170 μ L of 1M ammonium bicarbonate.

The released N-glycans were labelled with 2-aminobenzamide (2-AB) and were then purified from a mixture of 100% acetonitrile and ultrapure water in a 1:1 ratio (v/v). This was then analyzed by hydrophilic interaction liquid chromatography using ultra-performance liquid chromatography (HILIC-UPLC; Waters Corporation, Milford, MA) (Hou et al., 2019). As previously reported, the chromatograms were separated into 24 IgG glycan peaks (GPs) (Menni et al., 2018)." Please see the Method section (Page 15, lines 346-355).

(5) Additional statistical analyses should be performed, including multiple comparisons with p-value adjustment, false discovery rate (FDR) control, and Pearson correlation (line 291).

As suggested, we have performed additional statistical analyses and mentioned the results in the revised manuscript.

"Positive correlations were observed between the ASM subsets and both galactosylation ($p=0.017$, $r_s=0.418$) and sialylation ($p=0.008$, $r_s=0.458$) in the antibody Fc region, as well as between the PB subsets and sialylation ($p=0.036$, $r_s=0.372$) (Figure 4A-C). (Page 8, lines 180-183)"

"The Benjamini - Hochberg (BH) method was used to adjust the raw p-values from DEG analysis, controlling the false discovery rate (FDR)." Please see the Materials and Methods (Page 16, lines 369-371).

(6) Quality control should be conducted prior to the IgG N-glycome analysis. Additionally, both biological and technical replicates are essential to assess the reproducibility and robustness of the methods.

Thank you for the suggestion. We have added descriptions on the biological and technical replicates in the Method section.

"Our study incorporated both biological and technical replicates to ensure a robust glycomic profiling analysis. Specifically, we analyzed paired acute/convalescent-phase samples from 65 confirmed HFRS patients to assess inter-individual biological variability, while technical reproducibility was validated through comparison with standard chromatographic peak plots (Vučković et al., 2016). This dual-replicate strategy enabled a comprehensive evaluation of both biological heterogeneity and assay precision." (Page 15, lines 356-362).

(7) Multiple regression analysis should be conducted to evaluate the influence of genetic and environmental factors on the IgG N-glycome.

As suggested, we have conducted multiple regression analysis to evaluate the influence of genetic and environmental factors on the IgG N-glycome. These results have been provided in the revised Result section.

"Multivariate linear regression was employed to mitigate potential confounding by genetic and environmental factors in the glycomics analysis. While no significant associations were observed for most glycan models (fucosylation, $p=0.526$; bisecting GlcNAc, $p=0.069$; and sialylation, $p=0.058$), we discovered sex showed a potentially influential effect on galactosylation ($p=0.001$) (Supplementary files 5-8). These results suggest that while most glycan features appear unaffected by the examined covariates, galactosylation may be subject to sex-specific biological regulation." (Page 7, lines 153-160).

(8) Line 196. Additional discussions should be included, focusing on the underlying correlation between the differential expression of B-cell glycogenes and the dysregulated IgG N-glycome profile, as well as the potential molecular mechanisms of IgG N-glycosylation in the development of HFRS.

Thank you for your suggestions. We have added these contents in the Discussion section.

"Antibody-related glycogenes are significantly activated following Hantaan virus infection. We noted that ribophorin I and II (RPN1 and RPN2) were significantly upregulated in the ASM/IM/PB/RM subsets after Hantaan virus infection, which linked the high mannose oligosaccharides with asparagine residues found in the Asn-X-Ser/Thr consensus motif (Hwang et al., 2025). We speculate that they continuously attach the synthesized glycan chains to the constant region of antibodies during antibody synthesis. Similarly, fucosyltransferase 8 (FUT8) in the ASM subset, catalyzing the alpha1-2, alpha1-3, and alpha1-4

fucose addition (Wang & Ravetch, 2019; Yang et al., 2015), was downregulated in the mRNA translation, and the levels of fucosylated antibodies were naturally lower in the acute HFRS patients. Meanwhile, the beta-1,4-galactosyltransferase (beta4GalT) gene expression was significantly elevated in the ASM subpopulation during the acute phase, which also correlated with increased levels of galactosylated antibodies in serum (Wang & Ravetch, 2019). However, we did not observe significant upward changes in sialyltransferase mRNA expression in the acute HFRS patients, similar with the finding from severe COVID-19 cohorts (Haslund-Gourley et al., 2024). The neuraminidase 1 (NEU1) gene is strikingly upregulated and may potentially explain the decreased sialylation on the secreted HTNV-specific IgG antibodies during convalescence. Overall, the glycosylation of immunoglobulin G is regulated by a large network of B-cell glycogenes during HTNV infection." Please see the Discussion (Page 11, lines 254-273).

Reviewer #2 (Public review):

(1) While it is great to reference prior publications in the Materials and Methods section, the current level of detail is insufficient to clearly understand the study design and experimental procedures performed. Readers should not be expected to consult multiple previous papers to grasp the core methodological aspects of the present paper. For instance, the categorization of HFRS patients into different clinical subtypes/ courses, and the methods for measuring Fc glycosylation should be explicitly described in the Materials and Methods section of this manuscript.

Many thanks for your comments. We have added more details regarding the study design and experimental procedures in the Materials and Methods section. "Clinical specimens were collected from HFRS patients who were hospitalized in Baoji Central Hospital between October 2019 and January 2022. Patients were categorized into four clinical subtypes (mild, moderate, severe, and critical) based on the diagnostic criteria for HFRS issued by the Ministry of Health (Ma et al., 2015). This study was approved by the ethics committee of the Shandong First Medical University & Shandong Academy of Medical Sciences (R201937). Written informed consent was obtained from each participant or their guardians.

The clinical course of HFRS is grouped into acute (febrile, hypotensive, and oliguric stages) and convalescent (diuretic and convalescent stages) phases. The acute phase was defined as within 12 days of illness onset, and the convalescent phase was defined as a period of illness lasting 13 days or longer (Tang et al., 2019; Zhang et al., 2022). The earliest sample was selected if there were multiple blood samples available in the acute phase and the last available sample before discharge was selected if there were multiple blood samples in the convalescent phase.

Briefly, the diluted plasma samples were transferred onto a 96-well protein G monolithic plate (BIA Separations, Slovenia) for the isolation of IgG. The isolated IgG was eluted with 1 mL of 0.1 M formic acid and was immediately neutralized with 170 µL of 1M ammonium bicarbonate.

The released N-glycans were labelled with 2-aminobenzamide (2-AB) and were then purified from a mixture of 100% acetonitrile and ultrapure water in a 1:1 ratio (v/v). This was then analyzed by hydrophilic interaction liquid chromatography using ultra-performance liquid chromatography (HILIC-UPLC; Walters Corporation, Milford, MA) (Hou et al., 2019). As previously reported, the chromatograms were separated into 24 IgG glycan peaks (GPs) (Menni et al., 2018)." Please see the Materials and Methods (Page 13, lines 290-303, and Page 15, lines 346-355).

(2) The authors should explain the nature of their cohort in a bit more detail. While it appears that HFRS cases were identified based on IgM ELISA and/or PCR, these are

indicators of the Haantan virus infection. My understanding is that not all Haantan virus infections progress to HFRS. Thus, it is unclear whether all patients in the HFRS group actually had hemorrhagic fever. This distinction is critical for interpreting how the results observed relate to disease severity.

We are sincerely grateful for this valuable suggestion. We have carefully revised Figure 1 and the texts (Page 5, lines 104-107) in the revised manuscript.

"To characterize the humoral immune profiles in HFRS patients, we enrolled 166 suspected HTNV-infected patients who were admitted to Baoji Central Hospital in Shaanxi Province, China, between October 2019 and January 2022. Among them, 65 met the inclusion criteria and were included in the study (Figure 1)."

(3) The authors state that: "A 4-fold or greater increase in HTNV-NP-specific antibody titers usually indicates a protective humoral immune response during the acute phase", but they do not cite any references or provide any context that supports this claim. Given that in their own words, one of the most significant findings in the study is changes in glycosylation coinciding with this 4-fold increase, it is important to ground this claim in evidence. Without this, the use of a 4-fold threshold appears arbitrary and weakens the rationale for using this immune state as a proxy for protective immunity.

Thank you for the suggestion and we have provided relevant references in the Results section (Page 8, lines 171-173).

According to the Expert Consensus on Prevention and Treatment of Hemorrhagic Fever with Renal Syndrome (HFRS) (<https://ts-cms.jundaodsj.com/file/163823638693909.pdf>), a confirmed diagnosis requires, based on a suspected or clinical diagnosis, one of the following: positive serum-specific IgM antibodies, detection of Hantavirus RNA in patient specimens, a four-fold or greater rise in titer of serum-specific IgG antibodies in the convalescent phase compared to the acute phase, or isolation of Hantavirus from patient specimens. A four-fold or greater rise in titer of convalescent serum-specific IgG antibodies compared to the acute phase not only suggests a recent Hantaan virus infection, but also the production of antibodies helping to combat the viral infection. In addition, the antibody glycosylation modifications may thus play a significant role in the antiviral immune response.

(4) The authors also claim that changes in Fc glycosylation influence recovery from HFRS - a point even emphasized in the manuscript title. However, this conclusion is not well supported by the data for two main reasons. First, the authors appear to measure bulk IgG Fc glycans, not Fc glycans of Hantaan virus-specific antibodies. While reasonable, this is something that should be communicated in the manuscript. Hantaan virus-specific antibodies are likely a very small fraction of total circulating IgG antibodies (perhaps ~1%), even during acute infection. As a result, changes in bulk Fc glycosylation may (or may not) accurately reflect the glycosylation state of Hantaan virus-specific antibodies. Second, even if the bulk Fc glycan shifts do mirror those of Hantaan virus-specific antibodies, it remains unclear whether these changes causally drive recovery or are merely a consequence of the infection being resolved. Thus, while the differences in Fc glycosylation observed are interesting - and it is tempting to speculate on their functional significance - the manuscript treats the observed correlations as causal mechanistic insight without sufficient data or justification.

Thank you for your valuable comments. This study measured bulk IgG Fc glycans, not Fc glycans of Hantaan virus-specific antibodies. We have described this limitation in the Discussion section (Page 12, lines 274-280). As reported in previous studies (references provided below), the changed pattern of virus-specific IgG N-glycans may reflect the total IgG

N-glycome. Nevertheless, more studies are clearly needed to directly measure virus-specific IgGs and to clarify the causal mechanistic insights.

References

- (1) Mads Delbo Larsen, Erik L de Graaf, Myrthe E Sonneveld, et al. Afucosylated IgG characterizes enveloped viral responses and correlates with COVID-19 severity. *Science*. 2021 Feb 26;371(6532): eabc8378.
- (2) Chakraborty S, Gonzalez J, Edwards K, et al. Proinflammatory IgG Fc structures in patients with severe COVID-19. *Nat Immunol*. 2021 Jan;22(1):67-73.
- (3) Tea Petrović, Amrita Vijay, Frano Vučković, et al. IgG N-glycome changes during the course of severe COVID-19: An observational study. *EBioMedicine*. 2022 Jul ;81: 104101.
- (4) Hou H, Yang H, Liu P, et al. Profile of Immunoglobulin G N-Glycome in COVID-19 Patients: A Case-Control Study. *Front Immunol*. 2021 Sep 23;12: 748566.

(5) Fc glycosylation is known to be influenced by covariates such as age and sex. While it is helpful that the authors stratified the patients by age group and looked for significant differences in glycosylation across them, a more robust approach would be to directly control for these covariates in the statistical analysis - such as by using a linear mixed effects model, in which disease state (e.g., acute vs. convalescent), age, and sex are treated as fixed effects, and subject ID is included as a random effect to account for repeated measures. This would allow the authors to assess whether observed differences in Fc glycosylation remain significant after accounting for potential confounders. This could be important given that some of the reported differences are quite small, for example, 94.29% vs. 94.89% fucosylation.

Thank you for your valuable suggestion. As suggested, we have conducted multiple regression analysis to evaluate the influence of genetic and environmental factors on the IgG N-glycome, and have provided these results in the revised Result section.

"Multivariate linear regression was employed to mitigate potential confounding by genetic and environmental factors in the glycomics analysis. While no significant associations were observed for most glycan models (fucosylation, $p=0.526$; bisecting GlcNAc, $p=0.069$; and sialylation, $p=0.058$), we discovered sex showed a potentially influential effect on galactosylation ($p=0.001$) (Supplementary files 5-8). These results suggest that while most glycan features appear unaffected by the examined covariates, galactosylation may be subject to sex-specific biological regulation." (Page 7, lines 153-160).

(6) The manuscript states that there are limited studies on antibody glycosylation in the context of HFRS, but does not cite any relevant literature. If prior work exists, it should be cited to contextualize the current study. If no prior studies have been conducted/reported, to the author's knowledge, that should be stated explicitly to show the novelty of the work.

Thank you for your suggestion. To our knowledge, there has been no prior reports regarding the regulation of IgG glycosylation in HFRS, particularly in relation to seroconversion. We have reworded this sentence in the revised manuscript. "Importantly, there have not been prior studies specifically examining plasma IgG N-glycome profiles derived from chromatographic peak data in HFRS patients, particularly in relation to seroconversion status. This gap in our knowledge motivated our systematic investigation of both total and virus-specific IgG glycosylation dynamics during acute infection." Please see the Introduction (Page 5, lines 92-96).

Reviewer #2 (Recommendations for the authors):

Minor points:

(1) Line 47, 78: The use of the word 'However' appears to be an incorrect expression.

We have made this correction.

(2) Line 127: The term 'glycome' should be replaced with 'N-glycome,' and all relevant expressions should be corrected accordingly, such as 'N-glycosylation.'

We have made this correction.

(3) Line 84-87: The sentence 'A total of 166 HFRS patients...' contains a grammatical error.

We have made tis correction (Page 5, lines 99-101).

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